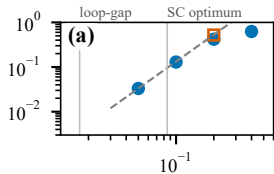
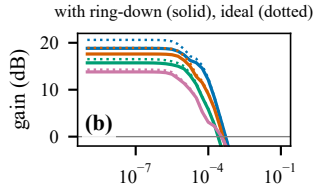


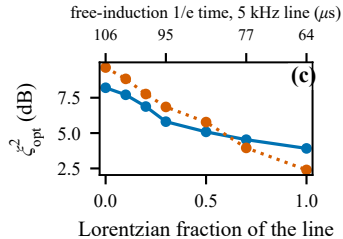
elimination loss (dB)



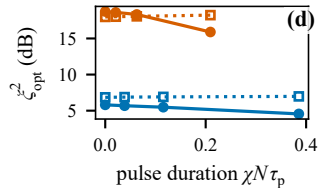
- exact, $N = 10$
- exact, $N = 20$
- $\propto (g\sqrt{N}/\Delta)^2$



- $\kappa/2\pi = 3 \text{ kHz}$
- $\kappa/2\pi = 10 \text{ kHz}$
- $\kappa/2\pi = 30 \text{ kHz}$
- $\kappa/2\pi = 100 \text{ kHz}$



- echo twist
- free twisting



- loop-gap, echo
- loop-gap, free
- SC operating point, echo
- SC operating point, free